Performance Dropped for PCIe3 Packet Switches Application Note

Introduction

Diodes' PCIe3 Packet Switches may encounter the performance drop issue under the specific conditions as following:

- When Endpoints use a large Maximum Read Request size for a single Memory Read Packet
- The packet size for most of TLP packets is smaller than the Maximum Payload Size.
- The Upstream port is x16 link width.

Diodes' PCIe3 Packet Switches include PI7C9X3G606GP-808GP-816GP-1224GP-1632GP-606GPQ-808GPQ-816GPQ-1632GPQ

When Endpoints use a large Maximum Read Request size for a single Memory Read Packet

Due to internal arbiter design, Diodes' PCIe3 Packet Switches may encounter the performance drop when Endpoints use a large Maximum Read Request size for a single Memory Read Packet. In such case, the PCIe bus will not release to other downstream ports until this Endpoint completes sending out all CPLD packets for this single Memory Read Packet. It may cause total performance drop when all downstream ports' Endpoints are running together, and some Endpoints take longer time to send out all CPLD packets for a single Memory Read Packet. It causes the PCIe Bus to underutilize because other Endpoints can't grab the PCIe Bus.

In order to avoid this performance drop issue, the Endpoint's driver can try a smaller Maximum Read Request size to decrease the time that Endpoints occupy the PCIe bus and allow other Endpoints to have chance to send out packets. It can improve the overall performance.

We have only seen this issue with Endpoints listed below when running 4 Endpoints at the same time.

Realtek RTL8125 2.5Gb NIC
Realtek RTL8126 5Gb NIC
Aquantia AQC107 10G NIC
Aquantia AQC113 10G NIC

Customers will need modify EP's device driver to try a smaller Maximum Read Request size to find the optimal setting for their application.

The packet size for most of TLP packets is smaller than the Maximum Payload Size

Diodes' PCIe3 Packet Switches have a fixed internal packet processing time. If the packet sizes for most of TLP packets are smaller than the Maximum Payload Size, the proportion of internal processing time is too high, resulting in a decrease in overall performance.

The Upstream port is x16 Link Width

When the upstream port links at GEN3x16 and the Maximum Payload size is 128 or 256 bytes, the performance is almost the same as that when the upstream port link is at GEN3x8. That is, if the root complex and all Endpoints can not support 512 byte Maximum Payload Size, it is advised not to set to 116 mode.

Below table is the performance simulation result. Please note they are simulation result and may be different us the real world.

Port Config	MPLS	Device -> RC	BW %	RC -> Device	BW %
Tile 0 G3 x16 (116) Tile 1 G3 4 X4 (416)	128	5.431213 GB/s	34.48%	7.372437 GB/s	46.80%
Tile 0 G3 x16 (116) Tile 1 G3 4 X4 (416)	256	7.743759 GB/s	49.15%	9.020996 GB/s	57.26%
Tile 0 G3 x16 (116) Tile 1 G3 4 X4 (416)	512	10.147095 GB/s	64.41%	9.705811 GB/s	61.61%
Tile 0 G3 x8 (216) Tile 1 G3 4 X4 (416)	128	5.253372 GB/s	66.69%	5.694275 GB/s	72.29%
Tile 0 G3 x8 (216) Tile 1 G3 4 X4 (416)	256	7.233124 GB/s	91.83%	7.231445 GB/s	91.81%
Tile 0 G3 x8 (216) Tile 1 G3 4 X4 (416)	512	7.326202 GB/s	93.01%	7.328186 GB/s	93.03%

MPLS: max payload size
Device->Host: packets direction is from device to root complex
Host->Device: packets direction is from root complex to device
BW%: Bandwidth usage.